India’s Semiconductors and Strategic Dependence: India’s Transition from Imports to Indigenous Manufacturing

Nishant
nishant12200333@davuniversity.org
MBA Student, DAV University &

Digital Media Executive, A-Sonne

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Abstract

The Semiconductor Industries are the foundation of modern digital economies, supporting a wide range of industries such as consumer electronics, telecommunications, automobile manufacture, defense systems, data infrastructures, and artificial intelligence (AI). even having a huge technical staff, powerful chip design capabilities, and a rapidly growing electronics market, India was absolutely missing from semiconductor production for decades and in result depending primarily on imports to meet domestic or homegrown demands. This paper analyses the fundamental economic and policy issues that contributed to India's slow entry into semiconductor manufacturing, such as high capital expenditure, technological complexity, infrastructure limitations, scattered supply chains, and delayed government action in past.

The research evaluates India's semiconductor consumption patterns, industry wise demand distribution, and import-export patterns to identify the scope and its effect of its global dependencies. The analysis highlights India's dependencies on global manufacturing hubs such as Taiwan, South Korea, and the Netherlands, and places it within the larger context of the global semiconductor value chain. The study uses secondary data from official government publications, Union Budget 2026-2027 official documents, industry reports, and policy announcements to evaluate India's changing semiconductor strategy and recent progress toward indigenous capacity development.

The focus is on India's developing manufacturing and design activities, such as new fabrication and ATMP facilities, academic-industry collaboration, and indigenous chip development projects like the Vikram-32 processor. The study looks more closely into the Indian government's renewed policy focus, particularly the fiscal and financial backing outlined in the Union Budget 2026-27, as well as the expansion of the India Semiconductor Mission (ISM), both of which aim to reduce import dependence, strengthen supply-chain resilience, and build a sustainable domestic semiconductor ecosystem.

By Using trade data, evaluation of policy, and recent developments in technology, the report presents an in-depth study of India's journey from semiconductor import dependence to its own manufacturing potential. The analysis is backed up by expert opinions, industry

assessments, and forward-looking expectations from senior policymakers, industry leaders, and research institutions, which provide insights into expected growth patterns, emerging opportunities, and challenges that will shape the sector's long-term growth.

Keywords: Semiconductor manufacturing, strategic dependence, import–export dynamics, India Semiconductor Mission, indigenous chip development, global semiconductor value chain, Union Budget 2026–27, and supply-chain resilience.

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1 Introduction

1.1 Background of the Global Semiconductor Industry.

The global Business of semiconductor functioning as the foundation for the current technological revolution, supporting electronic devices and systems to work in almost every sector. Semiconductors play key roles in consumer electronics, telecommunications, automotive systems, industrial automation, defense technologies, data centers, and new domains like artificial intelligence, machine learning, and the Internet of Things. Their function has shifted from simple logic and memory applications to highly complex nanotechnology integrated circuits which boost computational performance, energy efficiency, and technological advancement. (PWC, 2024)

Since the mid-20th century, the semiconductor industry has seen rapid technical advancement, including constant shrinking in size and increasing the chip complexity, and a continuous improvement in processing power. This evolution has been driven by Continues research and development with capital-intensive manufacturing processes, and highly speeded global supply chains. Today, semiconductor production is divided into various stages, including chip design, silicon fabrication, assembly, testing, and packaging, each of which requires professional expertise and advanced infrastructure with significant capital investment. As a result, the industry has become one of the most capital-intensive and technologically Complex sectors in the global economy. (Semiconductor Industry Association, 2025)

In recent years, the importance of semiconductors in the industries has increased significantly due to digital transformation, supply-chain disruptions and expansion, and rising geopolitical tensions. The world now recognizes semiconductors not as commercial products but as Important assets essential for economic Development of a country, As a result, semiconductor policy is recognized as an important component of technological and economic planning, with countries increasingly looking to strengthen domestic capabilities while reducing external dependencies. (Deloitte, 2026)

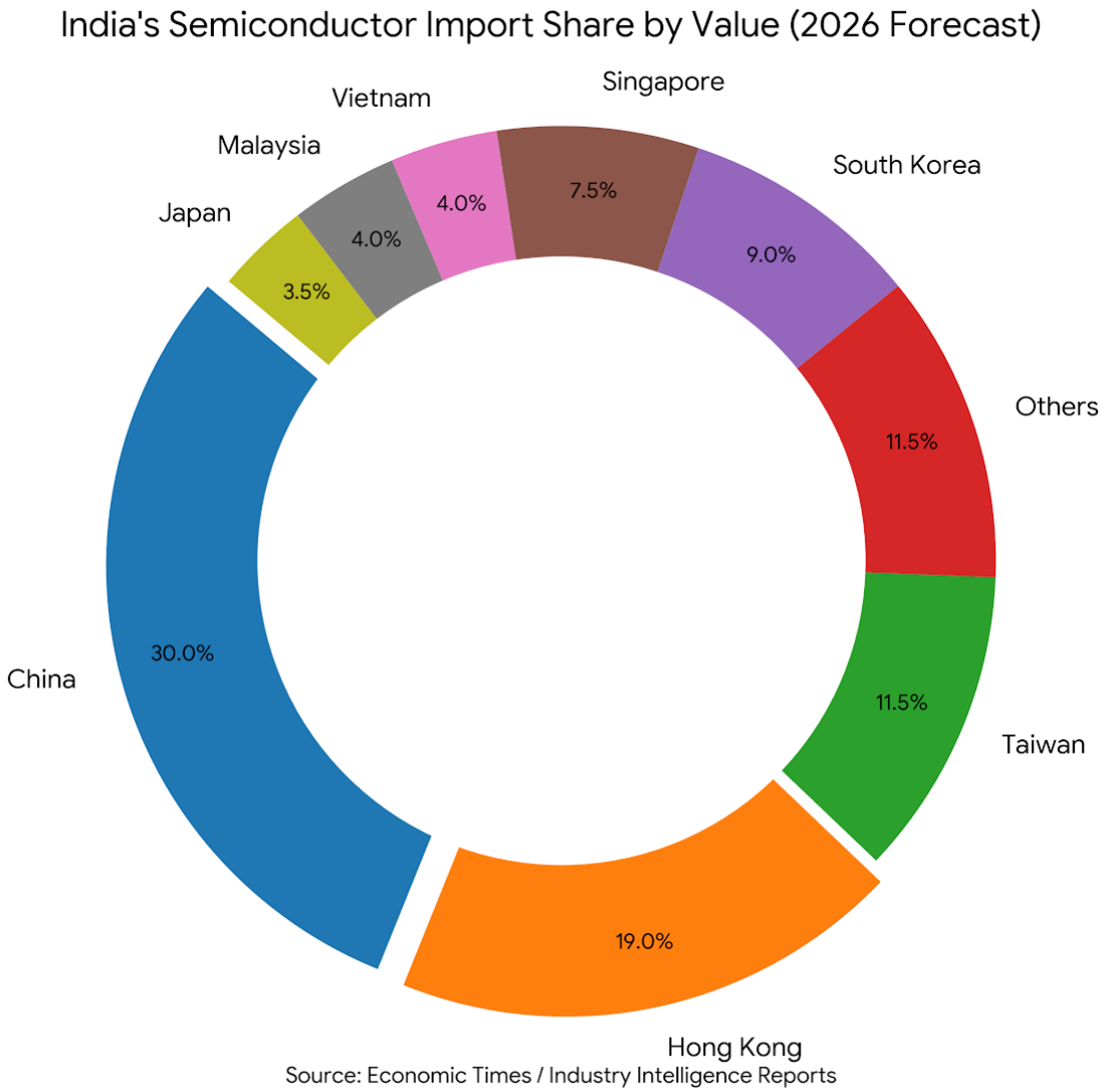
This global context serves as the foundation for analyzing India's position in the semiconductor Industry, especially when India's past absence in manufacturing of the Semiconductors, and dependency on other countries like China, Taiwan, South Korea, Malaysia's imports, and now India is Preparing to step into the global semiconductors manufacturing chain.

1.2 India’s Strategic Dependence on Semiconductors

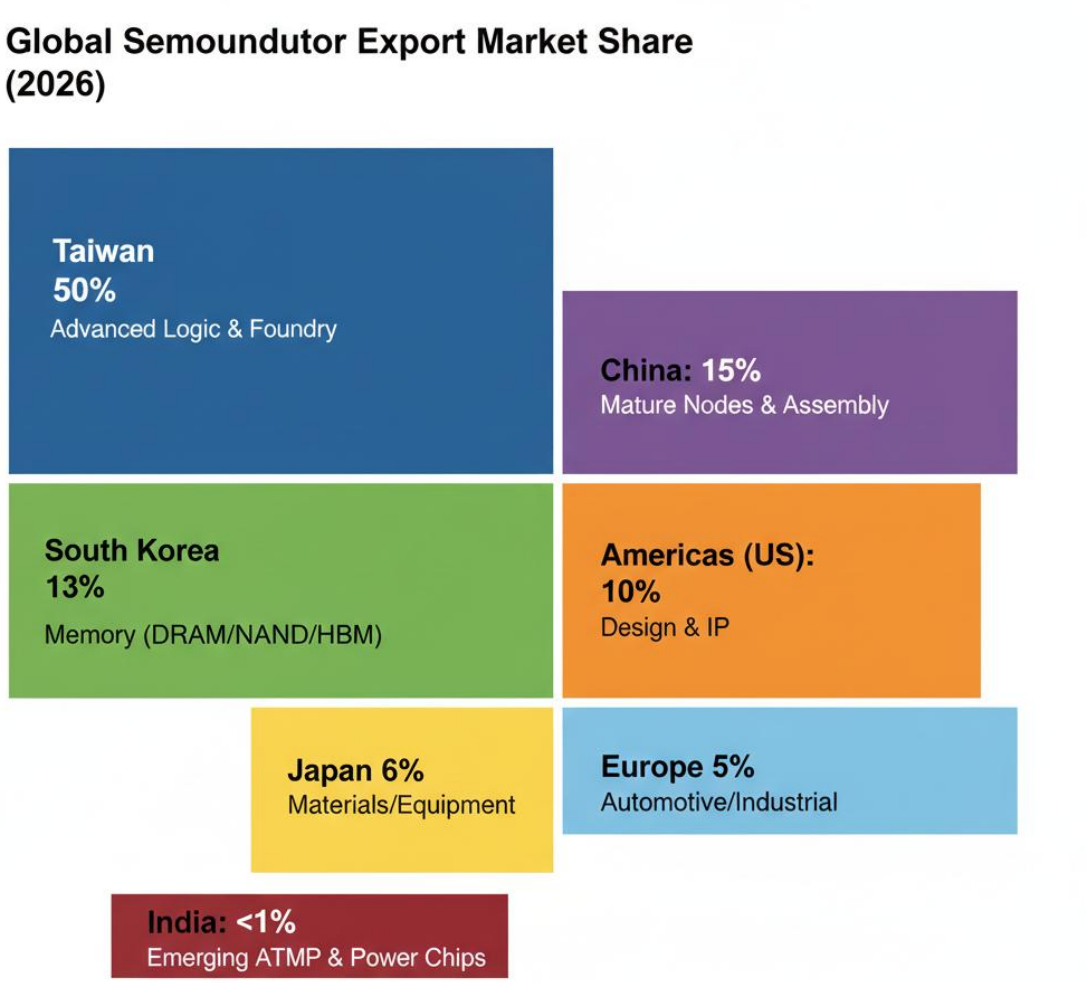
Historically, India's semiconductor ecosystem has had limited domestic fabrication capability and depends primarily on imported chips and components to meet fast rising domestic demand. Despite its strengths in software engineering and chip design services, India continues to import a significant portion of its semiconductors, mostly from a small number of East Asian manufacturing hubs such as China, Taiwan, South Korea, Japan, and Malaysia. According to official government evaluations, local production is now focused on assembling, testing, and packaging, with no expertise in semiconductor manufacturing, particularly at the commercial level. (Ministry of Electronics and Information Technology., 2025)

According to trade data and industry estimates, semiconductors are one of India's top electronics imports, exposing the economy to interruptions in the supply chain, price swings, and geopolitical risk. This dependence reflects the global nature of the semiconductor value chain, where advanced production is geographically concentrated and capital-intensive. (Volza Import Trade Data., 2025) Shipment-level trade intelligence demonstrates the scope and diversity of India's semiconductor imports, highlighting the country's limited domestic production base and reliance on foreign suppliers for both low-value individual parts and valuable integrated circuits.

As a result, semiconductor dependence has emerged not merely as a trade issue but as a strategic vulnerability affecting India’s electronics manufacturing, automotive production, digital infrastructure, and national security, thereby motivating recent policy interventions aimed at developing indigenous manufacturing capability.



(The Economic Times, 2026)



(WSTS, 2026)

1.3 Objectives and Scope of the Study

This study aims to provide a comprehensive and evidence-based evaluation of India's structural dependency on imported of the semiconductors or chip manufacturing, as well as its current transition to indigenous manufacturing of semiconductors within the global chip economy. The research aims to explain India's chronic lack of domestic fabrication capability by positioning it within broader international and historical contexts, despite the country's growing its electronics market, developing design capabilities, and big technical workforce.

The specific objectives of this study are to:

(i) critically examine the economic, technological, and historical elements that limited the growth of semiconductor production in India, paying special attention to infrastructural preparedness, capital intensity, technological entry hurdles, and postponed policy action.

(ii) measure India's import-export trends, sector-specific consumption, and semiconductor demand to evaluate the extent and strategic ramifications of external dependence.

(iii) research how the global semiconductor value chain is organized and how leading global manufacturing centers and technological innovators influence access to vital capabilities.

(iv) evaluate recent fiscal commitments and policy initiatives by the Indian government, including budgetary allocations and targeted incentive programs, in context of supply-chain resilience and industrial strategy and

(v) Evaluate India's present state of indigenous capabilities by examining new fabrication projects, modern packaging facilities, and industry-academia partnerships.

The scope of the study is focused on semiconductor manufacturing and its corresponding upward and downward relationships, which include fabrication, assembly, testing, packaging, and strategic design capabilities. The analysis is primarily based on secondary data from official government publications, budget documents, industry reports, and policy statements, which are supplemented by primary data gathered through a structured survey of engineering students, faculty members, researchers, and early-career professionals. While foreign comparisons are used to contextualize India's position in the global semiconductor ecosystem, the study is primarily concerned with India's policy issues, economic trade-offs, and long-term strategic potential. The study does not intend to give firm-level performance analysis, but rather evaluates India's semiconductor objectives from a macroeconomic and policy viewpoint.

2 Evolution of India’s Semiconductor Industry

India’s semiconductor landscape has evolved unevenly, shaped by historical industrial priorities, capital constraints, and limited integration into global manufacturing networks. While now the country acquired capabilities in engineering and talent and chip design, large-scale semiconductor production was unavailable for decades. The resulting imbalance created structural import dependency and slowed the development of a domestic industrial ecosystem. However, recent global shocks and strategic reassessments have pushed semiconductors to become one of India's industrial policy priorities.

2.1 Why India Missed Early Semiconductor Manufacturing

India's delayed entry into semiconductor production was the result of a complicated combination of economic, technological and policy issues, rather than a single strategic failure. During the early years of the global semiconductor industry, semiconductor production evolved as a highly capital-intensive sector that involved continual technological advancement and long-term government commitment. During this time, India's industrial strategy focused on self-reliance in core industries and public-sector manufacturing, leaving little fiscal and institutional opportunity to make long-term investment in high-risk technologies like semiconductor production. (Ministry of Electronic and Information Technology., 2025)

Capital constraints played an important role. Establishing and modernizing fabrication facilities requires investments that were difficult to rationalize under India's development framework, which prioritized infrastructure development, poverty alleviation, and economic stability etc. Limited access

to foreign funding and technology limits India's capacity to participate substantially in this advanced and complex industry. In additionally, inadequate collaboration with global semiconductor supply chains restricted opportunities for technological learning and scale-based cost reduction.

Lower Government support compounded these difficulties. Unlike other economies that pursued coordinated, export-oriented semiconductor strategies, India lacked an organized industrial framework that connected government policy, private business, research institutes, and global markets. As a result, early initiatives remained isolated and technologically obsolete, limiting the development of a competitive manufacturing ecosystem

This historical pattern caused India to focus on downstream operations such as software services and chip design, while fabrication and advanced packaging were mostly outsourced. The legacy of this delayed manufacturing entry continues to affect India's semiconductor strategy, emphasizing the significance of recent legislation changes targeted at closing long-standing structural gaps.

2.2 Capital, Technology, and Infrastructure Constraints

Semiconductor and chip manufacturing industry has some of the toughest entry barriers among modern industrial sectors, and these constraints played a significant role in limiting India's initial participation. Fabrication facilities require a significant initial financial investment, extended return periods, and ongoing reinvestment to keep up with rapid technological innovation. For an economy balancing numerous developmental goals, such capital intensity greatly restricted the viability of continuous investment in chip manufacture in previous decades.

Technological constraints compounded capital issues. Advanced semiconductor manufacture requires access of innovative process technologies, specialized equipment, and deep manufacturing expertise gained via countless production cycles. India's limited access to these global technological networks limited prospects for process learning and innovation. Furthermore, prior geopolitical periods' export restrictions and restricted technology transfers hampered India's capacity to acquire advanced industrial capabilities, strengthening technological dependence on foreign suppliers.

Infrastructure issues were also identified as a key obstacle. Semiconductor manufacturing involves extremely reliable energy supplies, ultra-pure water systems, smart logistics, and specialized industrial ecosystems. Historically, India lacked the interconnected infrastructure clusters needed to enable highly precise manufacturing at scale. The lack of mature supplier networks for materials, chemicals, and equipment raised operating risks and costs, lowering the appeal of domestic manufacturing initiatives.

Together Capital scarcity, technology limitations, and lack of infrastructure all contributed to a feedback loop that discouraged private investment and slowed Semiconductors industry growth. These restrictions explain not just India's delayed entry into semiconductor manufacturing, but also the strategic importance of recent governmental interventions targeted at lowering entry barriers, reducing risks investment, and establishing supporting infrastructure for long-term competitiveness in manufacturing. (Ministry of Electronic and Information Technology., 2025)

2.3 Role of Government Intervention and Policy Delay

Government intervention has been an important characteristic of the global semiconductor sector, where long manufacturing periods, significant uncertainty, and strategic consequences frequently

prevent entirely market-driven investment. In India, however, governmental involvement in semiconductor manufacture came late and lacked the consistency that is essential for maintaining a marketplace. This delay was caused by broader development priorities, fiscal restrictions, and shifting views of technology risk, rather than policy indifference.

Throughout the post-independence period, India's policy priorities were oriented on food security, infrastructure development, public-sector expansion, and basic industrial capabilities. Semiconductor production, which provided uncertain short-term returns and required long-term financial and structural investment, did not naturally fit with these principles. As a result, early government initiatives in electronics and microelectronics were fragmented, small in scope, and insufficient for maintaining commercially viable fabrication facilities.

Additionally, government engagement in the semiconductor industry requires not just financial assistance, but also organizational capacity to handle complex industrial collaborations and changing technological requirements. India's governmental structure, particularly in earlier decades, was not completely prepared to handle the rapidity and complexity of semiconductor innovation. As a result, decision-making processes frequently lag behind global industrial cycles, compounding India's reliance on foreign technologies. (Economic Survey of India 2023–24., 2023)

In recent years, however, the political significance of semiconductors has become increasingly clear. Global supply-chain disruptions, increased international competition, and the expanding importance of digital technologies have caused a rethinking of the government's role in semiconductor and chip manufacturing. This shift indicates an evolution from previous caution, with government engagement viewed as a crucial motivator for addressing historical delays and facilitating India's transition to indigenous semiconductor capacity.

3 Macroeconomic Overview of India’s Semiconductor Market

Rapid economic growth, restructuring the economy, and increasing digital adoption all have an impact on India's semiconductor business. As the economy shifts toward greater value-added manufacturing and services, demand for semiconductors has grown across different sectors, making chips a critical input for both industrial productivity and consumer markets. Unlike in the early stages of India's industrialization, semiconductor demand is now driven not only by electronics consumption, but also by broader macroeconomic factors such as urbanization, higher disposable incomes, digital infrastructure expansion, and industry-wide technological upgrading. From the perspective of macroeconomics semiconductors serve as a supportive component rather than an independent industry. Growth in industries such as electronics manufacturing, automotive manufacture, telecommunications, defense, and data services has resulted in increased semiconductor usage. This demand-side increase is exceeding the growth of domestic manufacturing capacity, raising the dependency on imports and highlighting the crucial significance of supply stability and cost efficiency.

This section studies the macroeconomic variables that drive India's semiconductor demand, with a focus on domestic consumption trends, sector-specific usage patterns, and the relationship between economic growth, digitalization, and semiconductor intensity. The analysis highlights why semiconductor demand has become an important component of India's industrial and technology policy agenda by positioning it within the country's broader economic trajectories.

3.1 Sector-Wise Consumption of Semiconductors in India

Semiconductor usage in India is spread over a wide range of sectors, showing the growing importance of electronics in the country's economic activities. Unlike earlier phases, when demand was mainly for consumer electronics, semiconductors are now used in a variety of industries with different performance, reliability, and security needs. This diversification has increased the quantity and complexity of India's semiconductor needs.

Mobile phones and portable electronic devices consume the majority of semiconductors in India, accounting for over 35% of total demand. This supremacy is fueled by India's massive smartphone user base, rapid device replacement cycles, and ongoing rise in mobile data usage. This segment's semiconductor usage consists mostly of application processors, memory chips, radio-frequency components, and power management integrated circuits.

The IT and computing sector have the second-largest share, accounting for roughly 18% of semiconductor usage. This category contains semiconductors found in laptops, desktop computers, servers, data centers, and networking equipment. The growth of cloud services, digital platforms, and enterprise IT infrastructure has greatly raised demand for logic chips, memory devices, and networking processors in this field. (Indian Electronics and Semiconductor Association (IESA), n.d.)

Consumer electronics accounts for about 15% of semiconductor consumption. This category includes smart televisions, wearables, household appliances, and other digitally enabled consumer devices. The growing popularity of smart and connected goods has increased semiconductor intensity in domestic electronics, contributing to steady development in this area.

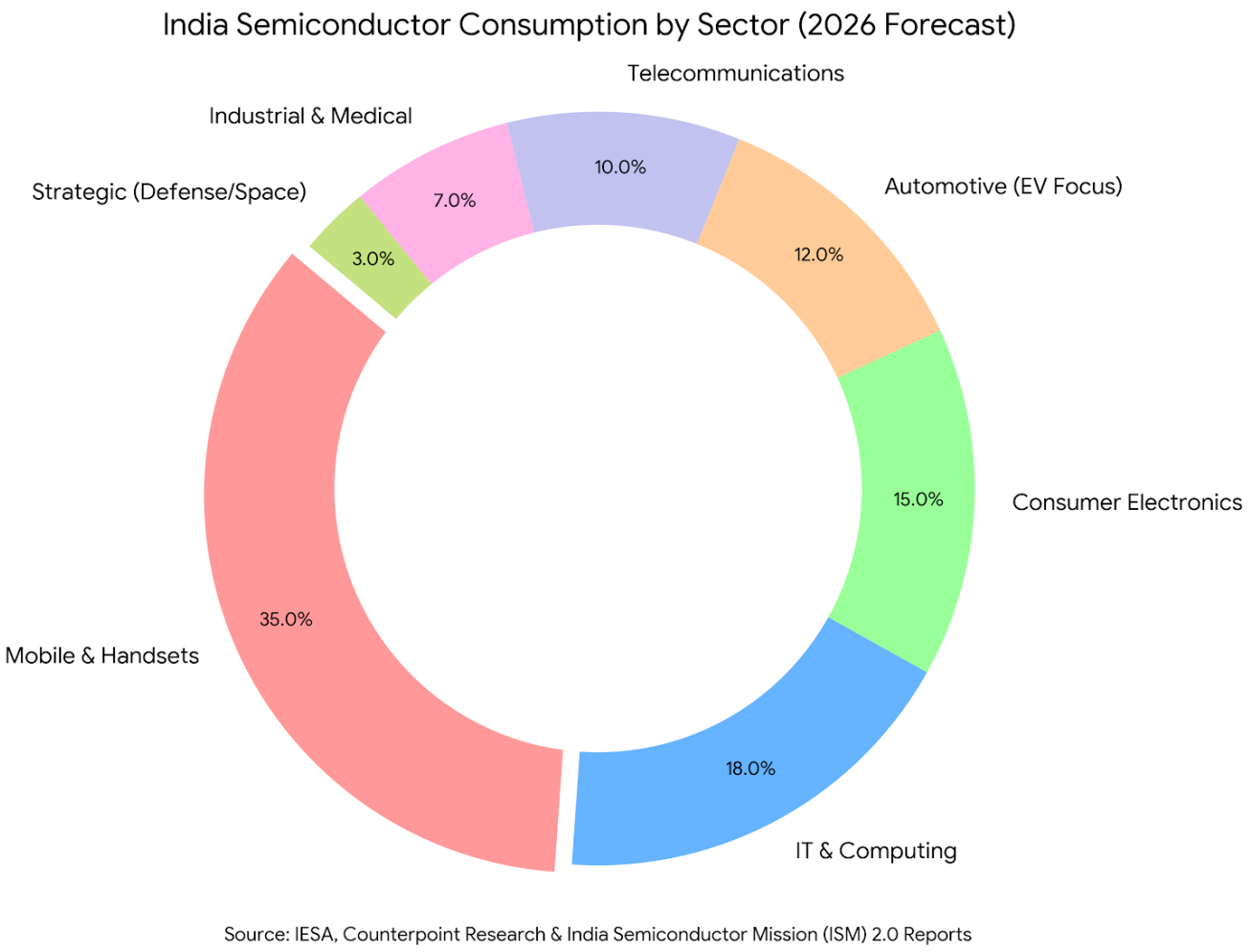
The automobile industry, with its focus on electric vehicles, comprises around 12% of semiconductor usage. Rising electronic content per car, fueled by power electronics, battery management systems, entertainment units, and improved driver-assistance systems, has boosted semiconductor demand in this segment. The automobile industry is likely to be one of the fastest-growing contributors to semiconductor use in the future years.

The telecommunications sector provides for around 10% of semiconductor demand, especially investment in mobile network infrastructure, broadband growth, and data transmission equipment. The deployment of modern communication technologies has boosted the demand for specialist semiconductors including radio-frequency chips, baseband processors, and optical components.

Industrial and medical electronics account for an estimated 7% of semiconductor consumption. This area includes semiconductors for industrial automation, medical devices, sensors, and control systems. Although smaller in volume than consumer categories, it stands out by superior reliability and longer product life cycles. Finally, key industries like defense and space account for roughly 3% of semiconductor demand. Despite its small size, this market is critical to strategic success because it relies on safe, reliable, and specialized semiconductor components.

Figure X represents the estimated sector-wise distribution of semiconductor consumption in India, highlighting the dominance of mobile and computer devices as well as the growing relevance of automobiles, telecommunications, and industrial applications. The industrial makeup highlights the range of semiconductor dependence in the Indian economy, highlighting its importance for diversified

domestic manufacturing capabilities.



(Indian Electronics and Semiconductor Association (IESA), 2025)

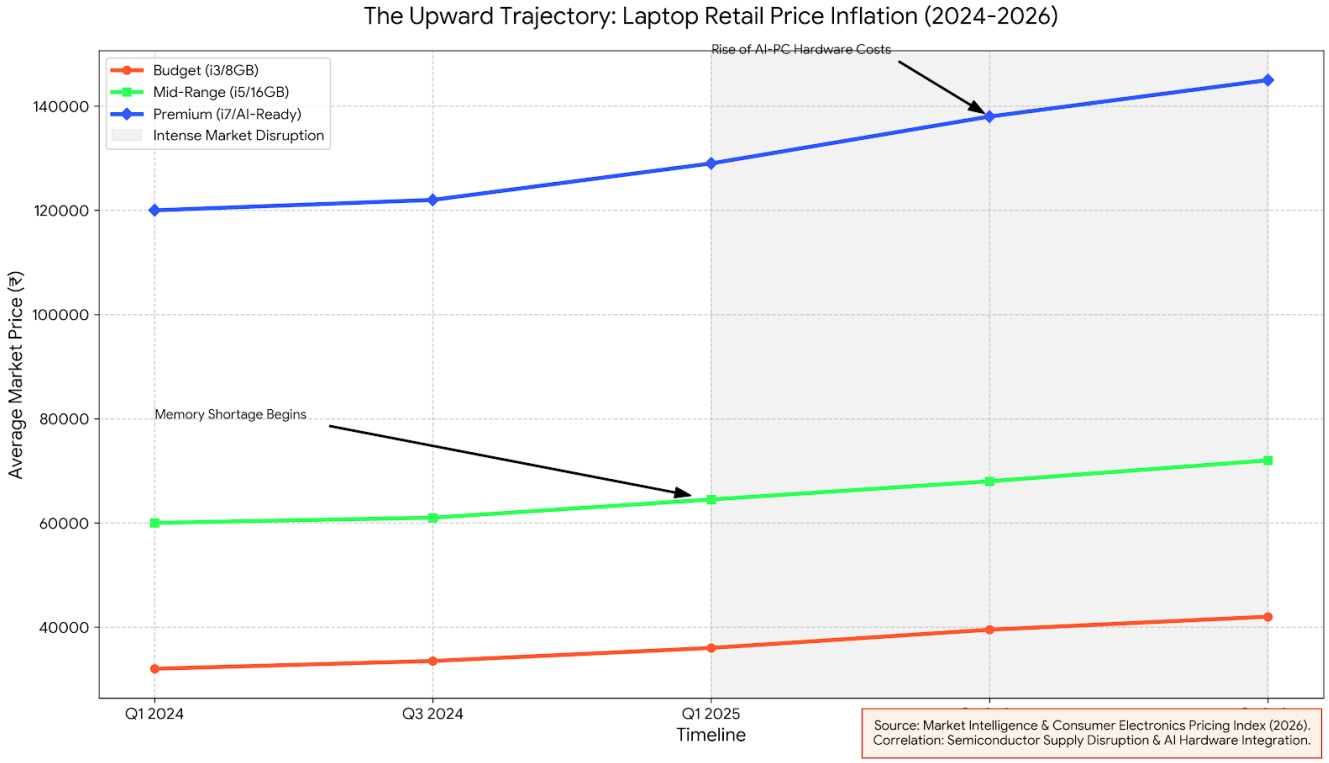
4 Impact of Global Supply Disruptions on India

The Global semiconductor supply disruptions over the last few years, particularly during the COVID-19 pandemic and afterwards time of increased geopolitical tension, have exposed fundamental weaknesses in India's technology and industrial environment. The global semiconductor supply chain, which is characterized by high geographic concentration and immediate production processes, proven extremely vulnerable to shocks caused by manufacturing shutdowns, logistical bottlenecks, export restrictions, and diplomatic tensions. For India, which relies largely on imported semiconductors to meet domestic demand, these interruptions had immediate and severe economic effects.

One of the most visible effects was seen in the automotive and electronics industries, where shortages of important chips caused production slowdowns, delayed product launches, and higher input costs

making the overall cost higher Automotive manufacturers had occasional factory shutdowns due to a lack of microcontrollers and power management integrated circuits, while consumer electronics manufacturers faced shortages in supplies that impacted inventory planning and market availability. Similar problems were reported in telecommunications equipment, industrial automation, and data processing technology, hurting both local output and export commitments. ***(Source Ability., 2025)***

Global supply imbalances also caused major price swings in high-performance computer and memory components. Strong demand for artificial intelligence accelerators and data center gear, fueled mostly by businesses like NVIDIA, has reduced global availability of advanced logic chips. Simultaneously, memory production restrictions resulted in significant increases in DRAM and RAM prices, thereby affecting the cost of laptops, cellphones, servers, and other electronic products in India. These price hikes were passed on to consumers and businesses, rising overall electronics costs and emphasizing India's poor ability to withstand global semiconductor pricing fluctuations.



With short-term shortages and high pricing effects the global disruptions exposed more serious issues of strategy for India. Dependence on a small number of offshore industrial hubs, particularly in East Asia, limited supply chain flexibility and exposed domestic companies to external policy decisions, such as export limits and technology controls. The lack of domestic manufacture capacity, as well as minimal storage or buffering mechanisms, increased India's weakness during times of global stress.

The consequences went beyond the business sectors to national security and key infrastructure. External dependence posed an increased danger in defense, space, and strategic communications, where semiconductor availability is tightly linked to operational capability. This experience strengthened the view of semiconductors as a strategic resource rather than a market-driven commodity.

Collectively, these events represent an important point in India's semiconductor policy discussion. Rising chip and memory prices, combined with supply uncertainty in vital industries, contributed to the government's awareness of the importance of domestic manufacturing competence, supply chain diversification, and ecosystem resilience. The global supply shock thus acted as a motivator for India's following policy interventions, such as mission-mode semiconductor efforts and large-scale budgetary support, with the goal of eliminating structural dependency and increasing long-term industrial stability.

5 India's Semiconductor Manufacturing Shift

After decades of being dependent primarily on foreign semiconductors, India has entered a new phase in its scientific and economic development. Rising domestic demand, global supply-chain disruptions, and strategic weaknesses have forced India to reconsider its long-standing dependency on external semiconductor manufacturing hubs. In response, the Indian government has launched a coordinated push to create indigenous semiconductor capabilities, which represents an important shift from previous, fragmented efforts.

This trend is seen in both early indigenous chip development efforts and large-scale governmental and budgetary initiatives aimed at establishing a comprehensive semiconductor ecosystem. Together, these developments indicate that semiconductor production in India is moving from a long-term aspiration to an actively pursued national goal.

5.1 Indigenous Semiconductor Development: The Vikram-32 Milestone (2025)



A turning point moment of India's semiconductor journey occurred in 2025, when the country showed its first totally domestic, mission-validated semiconductor processor. This development represented a transition from conceptual capability to practical evidence, adding credibility to India's semiconductor ambitions.

At SEMICON India 2025, India's first Made-in-India microprocessor, Vikram-32 (Vikram 3201), was formally given to the Prime Minister, representing a national milestone in semiconductor self-reliance. Vikram-32 is a 32-bit space-grade microprocessor developed by the Semiconductor Laboratory (SCL) of the Indian Space Research Organization. It is meant to work reliably despite the harsh heat, radiation, and mechanical conditions involved with space launches. (Ministry of Electronic and Information Technology., 2025)



Crucially, the processor was not restricted to laboratory validation. The first batch of Vikram-3201 devices was successfully flight-tested during the PSLV-C60 mission, demonstrating their functional reliability in real-world mission conditions. This in-orbit validation sets Vikram-32 apart from previous experimental attempts and establishes it as India's first operationally proved homegrown semiconductors.

The Vikram-32 was manufactured and packed at SCL Mohali using a 180 nm CMOS process, demonstrating India's current capacity in advanced-node semiconductor production. While this manufacturing node does not compete with advanced commercial logic technologies, it is still highly significant in space, defense, and strategic electronics where reliability, durability, and predictable performance take priority over transistor density.

Beyond space applications, Vikram-32's architectural reliability indicates possible applications in defense, aircraft, automobile electronics, and energy systems. Its progress highlights the benefit of

continued public R&D investment and institutional continuity following the beginning of the Mission ISM.

In addition to Vikram-32, academic and research institutes, including the National Institute of Technology Rourkela, presented indigenous encryption core ICs and sensor-based chips for secure communications, IoT, biomedical, and smart sensing applications. Despite their low scale, these efforts mark India's first end-to-end indigenous semiconductor manufacture.

These achievements are important not just because of their direct influence on import dependency, but also because they establish technological credibility, manufacturing learning, and system-level validation. The Vikram-32 milestone shows that India has passed a significant threshold from being a large semiconductor consumer to becoming a verified producer of indigenous chips.

5.2 New Fabrication and ATMP Plants in India

India has begun to translate policy intentions into actual manufacturing infrastructure. In recent years, new semiconductor fabrication and ATMP (Assembly, Testing, Marking, and Packaging) operations have been approved and started, with support from the national and state governments through the India Semiconductor Mission. (Blackridge Research and Consulting., 2025)



(Blackridge Research & Consulting, 2026)

These initiatives are intentionally spread across various states to boost regional manufacturing hubs and supply chain resilience. The emphasis on ATMP facilities reflects a phased manufacturing strategy, which prioritizes assembly and packaging as entry points while gradually expanding fabrication capability. Such facilities not only minimize import dependence, but also create jobs and facilitate integration into global semiconductor value chains.

Although most projects are still under construction or in the early phases of operation, they mark a significant transition from policy planning to implementation. When paired with indigenous design efforts and academic-industry collaboration, these manufacturing activities serve as the cornerstone for India's long-term semiconductor strategy.

Table 1: India's Major Semiconductor Projects (2025–26 Announced)

Company / Consortium	Location	Type of Facility	Capacity	Investment (INR Crore)
Tata Electronics & PSMC	Dholera, Gujarat	Fab (Foundry)	50,000 wafers/month	91,000
Micron Technology	Sanand, Gujarat	ATMP	101.6 million units/month	22,500
Tata Semiconductor Assembly	Morigaon, Assam	ATMP	48 million chips/day	27,000
Tower Semiconductor & Adani Group	Taloja, Mumbai	Fab (Foundry)	40,000 WSPM (Phase 1); total 80,000 WSPM	83,947
Renesas Electronics & Stars Microelectronics	Sanand, Gujarat	ATMP	15 million units/day	7,600
Kaynes Semicon	Sanand, Gujarat	ATMP	60 lakh chips/day	3,307
Suchi Semicon	Not announced	OSAT	Not announced	850
HCL–Foxconn JV	Jewar, Uttar Pradesh	OSAT	20,000 wafers/month; 36 million display driver chips/month	3,706

Table 2: India's Semiconductor Manufacturing Projects (2025 – Announced)

Company	Location of Plant	Type of Plant	Capacity (Annual)	Investment (INR Crore)
SiCSem Pvt. Ltd.	Info Valley, Bhubaneswar (Odisha)	Compound semiconductor fab (SiC)	60,000 wafers; 96 million packaged units	2,066
3D Glass Solutions Inc. (3DGS)	Info Valley, Bhubaneswar	Advanced packaging (3D Heterogeneous Integration	69,600 glass panels; 50 million units;	1,943

Company	Location of Plant	Type of Plant	Capacity (Annual)	Investment (INR Crore)
	(Odisha)	on glass substrate)	13,200 3DHI modules	
Advanced System in Package (ASIP) Technologies	Andhra Pradesh	Semiconductor packaging unit	96 million units	468
Continental Device India Ltd. (CDIL)	Mohali, Punjab	Discrete semiconductor expansion (Si & SiC devices)	158.38 million units	117

(Blackridge Research & Consulting, 2026)

5.3 Union Budget 2026–27 and the India Semiconductor Mission

Building on early indigenous semiconductor demonstrations like the Vikram-32 processor, the Indian government institutionalized its semiconductor goals with a solid policy and fiscal framework in the Union Budget 2026-27, which was presented on February 1, 2026. The budget represents a clear shift from exploratory projects to a planned, long-term strategy for improving India's competitiveness in manufacturing.

The budget includes semiconductor manufacturing as part of a larger effort to increase manufacturing in seven strategic and innovative sectors, highlighting the government's recognition of semiconductors as a critical input for modern economic growth, digital infrastructure, and national security. Within this model, semiconductor development is positioned not as a single industrial activity, but as an enabling pillar for several subsequent sectors of the economy. ***(Ministry of Finance, 2025)***

The budget includes a key announcement regarding the beginning of India Semiconductor Mission (ISM) 2.0, which broadens the scope of government participation beyond production incentives. According to the budget highlights, ISM 2.0 would prioritize semiconductor equipment and material production, full-stack Indian intellectual property creation, and domestic semiconductor supply chain strengthening. Importantly, the objective lays particular focus on industry-led research and training centers to assist technological development and the formation of a skilled workforce, addressing one of the structural gaps that have previously limited India's semiconductor goals.

In addition to ISM 2.0, the Union Budget 2026-27 increased the Electronics Components Manufacturing Scheme to ₹40,000 crore, providing significant support to the electronics manufacturing ecosystem. This expansion represents a policy recognition that semiconductor production cannot occur in isolation and requires a strong indigenous base of components, materials, and supporting industries. The strategy is intended to reduce dependence on imports, increase local value addition, and strengthen backward connections required for semiconductor manufacturing and assembling activities.

2. Modified Programme for Development of Semiconductor and Display Manufacturing Ecosystem in India (CS)

FINANCIAL OUTLAY (Rs in Cr)	OUTPUTS 2026-27			OUTCOME 2026-27		
2026-27	Output	Indicators	Targets 2026-27	Outcome	Indicators	Targets 2026-27
8,000	**a. Modified Scheme for setting up of Semiconductor Fabs in India**					
	1. Provide fiscal support on Project Cost for setting up of semiconductor fabs	1.1 Number of semiconductors fabs to be supported	1	1. Investments and employment generation in Semiconductor	1.1. Investment during the year under the Scheme (in Rs. crore)	4,000
					1.2. Employment generated by the supported units during the year under the Scheme (in numbers)	1,500
		1.2 Amount of fiscal support to be disbursed (in Rs. crore)	2,000			
	b. Modified Scheme for setting up of Compound Semiconductors / Silicon Photonics / Sensors Fab / Discrete Semiconductor Fabs and Semiconductor Assembly, Testing, Marking and Packaging (ATMP) / OSAT facilities in India					
	1. Provide fiscal support on Capital Expenditure for setting up of compound Semiconductors/ SiPh/ Sensors/ Discrete Semiconductors fabs / ATMP units	1.1 Number of compound / SiPh / Sensors/ Discrete Semiconductors fabs / ATMP units to be supported	9	1. Investments, production, exports and employment generation in Compound Semiconductor/ SiPh/ Sensors/ Discrete Semiconductor/ ATMP	1.1 Investment by units during the year under the Scheme (in Rs. crore)	11,000
		1.2 Amount of fiscal support to be disbursed (in Rs. crore)	5,000		1.2 Employment generated by supported units during the year under the Scheme (in numbers)	3,000
	c. Design Linked Incentive (DLI) Scheme					
	1. Provide fiscal	1.1 Number of Semiconductor	30	1. Design and	1.1 Number of	10

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Source: Ministry of Finance. (2026). Union Budget 2026-27 Expenditure Profile: Ministry of Electronics and IT. Available at indiabudget.gov.in

(Ministry of Finance, 2025)

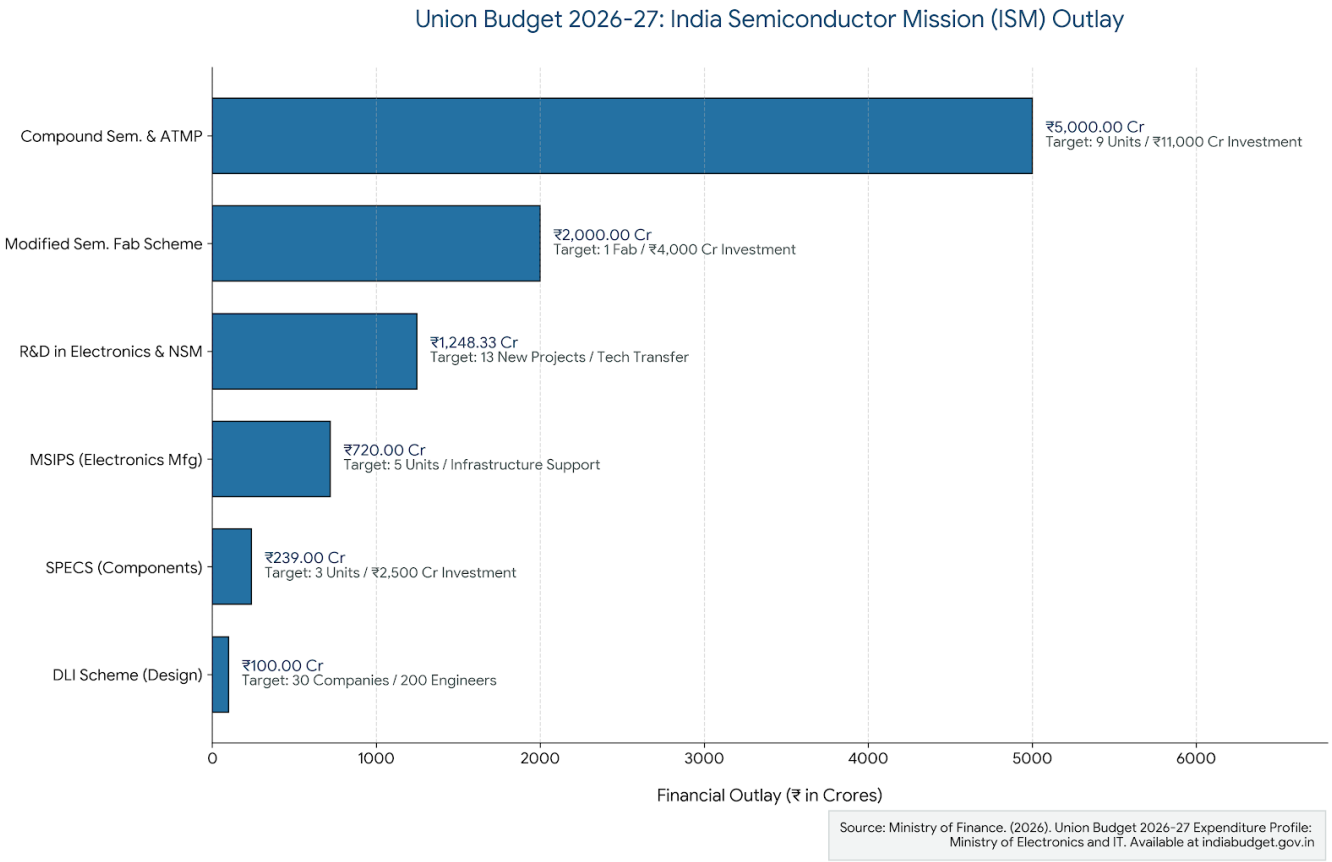
The budget's semiconductor priority is supported by associated programs targeted at boosting industrial infrastructure, capital goods capabilities, and supply chain stability. Measures like encouraging capital goods manufacture and promoting advanced industrial clusters are indirectly matched with semiconductor manufacturing needs, which require precise equipment, reliable logistics, and integrated industrial ecosystems.

FINANCIAL OUTLAY (Rs in Cr)	OUTPUTS 2026-27			OUTCOME 2026-27		
2026-27	Output	Indicators	Targets 2026-27	Outcome	Indicators	Targets 2026-27
					crore)	
	e. Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS)					
	1. Provide Fiscal Support to electronic components and semiconductors manufacturing units	1.1 Amount of disbursement done to units in FY 2026-27 (Rs. Crore)	239	1. Increased investment in electronic sector	1.1 Investment by units covered under the Scheme (in Rs. crore)	2,500
		1.2 No. of units supported	3	2. Increased production of electronic components and semi-conductors	2.1 Production by units covered under the scheme (in Rs. crore)	2,600
				3. Increased direct employment in electronics sector	3.1 Employment by units covered under the scheme (in number)	1,000

Source: Ministry of Finance. (2026). Union Budget 2026-27 Expenditure Profile:

Ministry of Electronics and IT. Available at indiabudget.gov.in
(Ministry of Finance, 2025)

Taken combined, the Union Budget 2026-27 and the establishment of ISM 2.0 mark a structural shift in India's semiconductor strategy. Unlike previous approaches that centered on fragmented monetary incentives, the 2026-27 framework employs an ecosystem-oriented strategy that integrates manufacturing, design, equipment, skills, and supply chains. While large-scale commercial semiconductor manufacture is still in its early stages, the budget shows that India has made a critical shift from managing semiconductor dependence to methodically developing local manufacturing capability.



(Ministry of Finance, 2025)

6. Comparative Analysis: India vs Leading Semiconductor Economies

A comparison of India to established semiconductor economies illustrates the scale of the difficulty India experiences in shifting from a consumption-driven market to a manufacturing-oriented ecosystem. While India has advantages in design skill and market demand, leading semiconductor countries have achieved dominance via decades of consistent investment, policy stability, and ecosystem depth.

Taiwan: Foundry-Led Global Dominance

Taiwan is the most advanced form of semiconductor production, with large-scale independent manufacturing facilities. Taiwan's success is built on early specialization, sustained government support, deep integration with global clients, and ongoing process innovation. Its manufacturing supremacy extends across advanced technological nodes, making it important to the global electronics and computing industries. In comparison, India is only now beginning to create fabrication capacity and is several technological generations behind.

South Korea: Vertically Integrated Manufacturing

South Korea uses a vertically integrated semiconductor approach, which includes design, fabrication, memory production, and downstream electronics manufacture. Strong coordination between the government and large industrial organizations has resulted in increased scale, capital intensity, and resilience between semiconductor cycles. India currently lacks such vertically integrated semiconductor manufacturing champions, instead relying on regulatory incentives to encourage both domestic and foreign participation.

China: State-Driven Capacity Expansion

China has set up a government directed semiconductor strategy centered on import substitution and capacity increase, notably in mature-node production, packaging, and design. Massive public investment and industrial policy support have supported fast capacity expansion, even though access to new technology remains restricted. India's method is more selective and fiscally responsible, but it is also slower to achieve size.

The Netherlands: Control of Critical Equipment

The Netherlands shows an alternative type of management to control over essential semiconductor production equipment rather than chip manufacture itself. By controlling advanced lithography equipment, the Netherlands holds a vital upstream position in the worldwide semiconductor value chain. This approach indicates that leadership may be obtained by specialization in enabling technologies, which India is only now exploring.

Overall, compared to these economies, India is still in the early stages of semiconductor manufacturing development. The disparity exists not only in technical nodes, but also in ecosystem maturity, supply-chain depth, capital intensity, and long-term industrial collaboration.

6.1 Key Takeaways for India from Global Semiconductor Leaders

The experience of leading semiconductor economies provides important lessons for India as it establishes its domestic manufacturing ecosystem.

1. Policy Continuity is critical

Semiconductor leadership necessitates decades of consistent, stable policy backing. Short-term incentives are insufficient without consistent regulatory and fiscal frameworks.

2. Scale matters as much as technology
Successful semiconductor hubs acquired competitiveness by rapidly increasing production capacity. India must transition from pilot initiatives to commercially viable, high-volume manufacturing.

3. Ecosystem Depth Is Essential
Fabrication factories cannot function in isolation. Equipment providers, material manufacturers, design firms, testing facilities, and shipping networks must all grow in tandem.

4. Strategic Segmentation Is Pragmatic
Fabrication factories cannot function in isolation. Equipment providers, material manufacturers, design firms, testing facilities, and shipping networks must all grow in tandem.

5. Strategic Segmentation is pragmatic
Competing instantly at the most sophisticated technological nodes may be unrealistic. Mature nodes, compound semiconductors, power electronics, and ATMP services all provide possible entry points.

6. Human Capital Must Match Infrastructure
Advanced manufacturing necessitates a continuous flow of experienced engineers, technicians, and researchers who are aligned with industry demands.

These lessons indicate that India's success will be determined less by imitating a single global model and more by modifying features that are appropriate for its economic structure, market size, and institutional capabilities.

7. Future Outlook and Growth Opportunities

India’s future in the semiconductor manufacturing sector will be determined by the combination of variables and Governmental policies consistency, ecosystem development, and evolving global market conditions. Although the country is still in the early stages of semiconductor production, recent institutional changes, budgetary support and technological initiatives have created many possibilities for India to gradually integrate into the global semiconductor value chain. The amount of this integration is going to be determined by India's capacity to transition from statement of policy to long-term and effective implementation in operations.

Aspect	**Baseline Scenario**	**Optimistic Scenario**	**Risk Scenario**
Policy Support	Stable, continued	Strong, well-executed	Weak, fragmented
Manufacturing Focus	ATMP + mature nodes	Fab + ATMP + design	Pilot-level only
Investment Flow	Moderate	High (domestic + FDI)	Low
Import Dependence	Gradual reduction	Sharp reduction	Remains high
Supply-Chain Resilience	Improving	Strong, integrated	Vulnerable

Aspect	Baseline Scenario	Optimistic Scenario	Risk Scenario
Employment Impact	Steady growth	High-skill job creation	Limited
Global Position	Packaging hub	Strategic manufacturing hub	Peripheral player

Scenario 1: Gradual Ecosystem Consolidation (baseline scenario)

According to the baseline scenario, India continues to enhance its position in semiconductor assembly, testing, and packaging (ATMP), while strategically growing fabrication capacity at mature technology nodes. The government's backing for the India Semiconductor Mission remains consistent, and domestic demand is growing in key areas such as automotive, consumer electronics, and industrial manufacturing. In this course, India gradually reduces its reliance on imports, strengthens its supply chain, and establishes itself as a reliable target for semiconductor packaging and related manufacturing activities, all without immediately competing in advanced nodes fabrication.

Scenario 2: Accelerated Manufacturing Scale-Up (Optimistic Scenario)

In a more optimistic scenario, efficient policy implementation, continued private investment, and more international collaboration will allow India to expand semiconductor production more quickly. Closer integration of chip design, production, and packaging appeals to global customers looking for ways to expand their supply chains. India can evolve as a strong competitor for power semiconductors, compound semiconductors, and customized logic circuits, resulting in increased local value and the development of skilled jobs. Under this scenario, India's role in the global semiconductor ecosystem shifts from a small player to a strategically important manufacturing and design manufacturing hub.

Scenario 3: Policy and Execution Constraints (Risk Scenario)

In a more optimistic scenario, efficient policy implementation, continued private investment, and more international collaboration will allow India to expand semiconductor production more quickly. Closer integration of chip design, production, and packaging appeals to global customers looking for ways to expand their supply chains. India can evolve as a strong competitor for power semiconductors, compound semiconductors, and customized logic circuits, resulting in increased local value and the development of skilled jobs. Under this scenario, India's role in the global semiconductor ecosystem shifts from a small player to a strategically important manufacturing and design manufacturing hub.

Scenario 3: Policy and Execution Constraints (Risk Scenario)

A negative outcome could occur if policy momentum weakens, the implementation of projects gets postponed, or coordination across the semiconductor ecosystem remains disorganized. In this circumstance, India's semiconductor initiatives may be limited to individual facilities and pilot programs, with a limited overall impact on import dependence and technological capabilities. Continued dependent on foreign suppliers would continue, and India's semiconductor goals may struggle to develop beyond superficial achievements.

Growth Opportunities Across the Value Chain

Regardless of the scenario, various potential for development remains fundamentally suitable in India's economic and technological environment. These include semiconductor design services, ATMP operations, power electronics, compound semiconductors, chips for automobiles and electric vehicles (EV's), and semiconductor equipment maintenance and support. India's large domestic market, rising digital infrastructure, and strong foundation of engineering expertise lay a good platform for growth in these categories, even if high-tech manufacturers remain difficult in the short-term period.

Long-Term Strategic Implications

In the long run, India's semiconductor growth will be dependent on continuous governmental cooperation, workforce development, and greater collaboration with global technological networks. Rather than hoping to copy the strategies of established semiconductor giants, India may design a unique strategy that blends scale, specialization, and strategic autonomy. If successfully implemented, India's semiconductor objectives can strengthen its industrial base, increase technical independence, and establish the country as a credible and powerful participant in the global semiconductor ecosystem.

8. Conclusion

This study analyzed India's position in the global semiconductor ecosystem by looking at its historical absence from semiconductor manufacturing, its current structural dependencies on imports, and the ongoing political push toward indigenous capability development. The analysis shows that India's delayed entry into semiconductor production was caused by a combination of capital constraints, technological limitations, infrastructure constraints, and delayed government involvement rather than a single policy failure. These forces have resulted in a widespread dependency on global industrial hubs, mainly in East Asia and Europe.

According to the macroeconomic assessment, semiconductor demand in India has grown fast across a wide range of industries, including consumer electronics, automobile manufacturing, telecommunications, industrial automation, defense, and data infrastructure. This broad-based demand growth has greatly increased India's vulnerabilities to global disruptions in the supply chain, price swings, and geopolitical issues, as highlighted by recent global semiconductor shortages and memory price increases. The impact of these disruptions confirmed the view of semiconductors as strategic assets rather than just economic components.

The research paper highlighted an important structural shift in India's semiconductor strategy, as illustrated by indigenous innovations in technology such as the Vikram-32 processor and the development of industry and academia collaboration in chip design and prototype development. While these activities are still limited in scale, they represent an important achievement India has progressed from theoretical participation to actual semiconductor production capabilities. The development of this technology has been strengthened by institutional and budgetary support from the India Semiconductor Mission and the Union Budget 2026-27, which combined represent an evolution toward ecosystem-oriented industrial strategy.

Comparative comparison with top semiconductor economies revealed that India is still in the early stages of manufacturing development, especially in terms of fabrication depth, supply-chain connection, and development. However, worldwide experiences indicate that leadership in the

semiconductor sector does not take a uniform path. India's developing approach, which focuses on ATMP services, advanced node fabrication, power and compound semiconductors, and design-led value creation, provides a pragmatic path that is consistent with the country's economic structure and market size.

The scenario-based research shows that India's future semiconductor direction will be strongly affected by policy stability, implementation efficiency, and ecosystem cooperation. Under ideal circumstances, India has the ability to considerably reduce import dependence, increase the resilience of its supply chain, and establish itself as a viable manufacturing and design hub in the global semiconductor industry. In contrast, poor execution or fragmented coordination can hinder progress and limit results to limited achievements.

at last India's semiconductor development has reached an important moment. Whereas significant gaps exist in comparison with global leaders, the combination of developing local demand, strategic government engagement, indigenous capability development, and global supply-chain transformations creates a historic opportunity. If continued with long-term commitment and coordinated execution, India's semiconductor programs have the potential to contribute significantly to technical independence, competitiveness in industry, and financial stability in the coming decades.

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